

Temperature in symbolic understanding

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Abstract

The same copular syntax that yields the tradition’s favorite truth condition, *snow is white*, yields the animal rights movement’s favorite slogan, *meat is murder*. We argue that the difference between the two lives in a single parameter of the *evaluation*, temperature, and that the term is not a metaphor: the parameter so named is literally the one that Harmonic Grammar and maximum entropy phonotactics fit to gradient acceptability data. The attention mechanism of transformers runs at the fixed temperature $1/\sqrt{d}$, and deployed language models expose the same parameter, applied one stage later to the choice of words, as the sampling knob. Grammaticality stars, analytic truths, thematic role assignment, anaphora resolution, and rhetorical force are readouts of one Gibbs competition at different operating points. We work through the mechanism on *meat is murder* and its political cousins *property is theft* and *taxation is theft*, and on anaphora resolution in *he beat him* discourses and Winograd schemas. A self-contained appendix develops the machinery – Gibbs distributions, annealing, energy landscapes, softmax, regularization – assuming nothing beyond linear algebra and one-variable calculus.

Keywords: temperature; Gibbs distribution; softmax; Harmonic Grammar; maximum entropy; Optimality Theory; gradience; generics; copular sentences; lexical semantics; rhetoric; slogans; anaphora resolution; Winograd schemas; annealing; attention; large language models

1 Introduction

Consider the minimal pair

- (1) a. Snow is white.
- b. Meat is murder.

By and large, any good syntactic parser of English will analyze (1a) and (1b) identically, as copulative sentences identifying the nominal (+N –V) constituent preceding the copula with the one following it. But (1a) is uncontroversial – the tradition’s very exemplar

of an unproblematic true proposition – whereas (1b) has a strong moral import, offering powerful rhetorical support for a vegetarian stance. The minimal pair is evidently not powered by syntax but by lexical information. The mainstream syntacto-semantic approach, which concludes, after a great deal of computation, that **snow**' $\subset$ **white**' and **meat**' $\subset$ **murder**', has very little traction over the key cases, even though it does something superficially interesting for at least three professors flunking at most seven students in more than one subject.

One standard response is to declare (1b) beyond the domain of grammar – metaphor, rhetoric, pragmatics. We hold instead with Wilks (1978): usage of this kind is the norm of ordinary language, not its periphery, and anyone who could not grasp such extensions could not be said to understand English properly. The other standard response accommodates each extension by a fresh posit – a second lexical entry, an additional sense, a type-shifting rule (see Kennedy 2011 for a careful execution of this strategy). We take the opposite, monosemic stance: if a single form is used, the burden of proof falls on those who wish to posit separate meanings (Ruhl, 1989).

The claim of this paper is that what separates the members of the pair, and quite generally what separates the literal from the extended, the categorical from the gradient, and the assertoric from the rhetorical, is a single parameter of evaluation which we call *temperature*. Underneath the claim runs a division we maintain throughout: composition builds *vectors* – a sentence, like a word, is a point in representation space – while *scalars* arise only in evaluation, so that traditional truth values are just the cold shadow of the sentence vector, not its meaning. Run cold over the lexicon as it is taught, (1a) is true and (1b) is simply false. But the falsity is manufactured at exactly one point: the restriction of *murder* to human victims. This restriction is installed by socialization, and the naive lexicon never had it – to the child, killing is killing, whatever the victim. Run cold over the naive lexicon, (1b) is as generically true as (1a). What the slogan asks for is the temporary lifting of that one restriction, and warmth is the mechanism by which a restriction can be lifted without being unlearned. Once it is lifted the rest is ordinary implication: if murder is to be avoided, as most people would instinctively agree, and meat is murder, it follows without any special stipulation that meat is to be avoided.¹

One point the paper insists on: *temperature* is meant literally. When a linguist hears the word applied to grammar, the natural assumption is that a suggestive analogy is being drawn from thermodynamics, to be cashed out, if at all, in some future theory. The assumption is wrong, and Section 2 together with Appendix A is designed to discharge it. The parameter has been in linguistics for decades under other names: Harmonic Grammar scores candidates by an energy function (Smolensky, 1986), Optimality Theory is its cold, strictly ranked limit (Prince and Smolensky, 1993), and maximum entropy phonotactics (Hayes and Wilson, 2008) is the Gibbs distribution, by that name, fitted to corpus data. The same parameter is set once, by the architecture, in the attention mechanism of the transformer (Vaswani et al., 2017), and exposed to every user of a deployed language

¹The syntactic framework in which this division of labor is embedded – a ‘bleached’ syntax whose entire contribution is a small set of linking instructions, *deltas*, over lexical statics in the sense of Kornai (2023) – is developed in a companion paper, *The bleaching of syntax*. Nothing below depends on its details beyond what is restated in place.

model as the sampling “temperature” – on the output side, as Appendix A.5 notes. One equation covers all of these, and the traditional symbolic computation of grammaticality – a derivation that returns a star or nothing – is its cold case, the limit in which the competition keeps only its best candidate, or the best few when they tie. The paper’s economy resides in the discipline this imposes: whatever cannot be put in the form of the one equation does not get to share the thermostat.

The paper is organized as follows. Section 2 introduces the one equation and its cold limit, where ungrammaticality stars and thematic role assignment live. Section 3 works through *meat is murder*: what runs cold, what requires warmth, and why the slogan is three words long. Section 4 extends the analysis to *property is theft* and *taxation is theft*, a pair chosen to show that the machinery measures rhetorical craft, not political merit. Section 5 turns to anaphora resolution – *he beat him* – and to Winograd schemas, arguing that the resolution is the same competition run over discourse. Section 6 collects the claims. The appendix is a tutorial for the working linguist, assuming nothing beyond linear algebra and one-variable calculus; readers to whom Gibbs distributions, annealing, and softmax are familiar can skip it, and readers to whom they are not may prefer to read it first.

2 One equation, run cold

Suppose a system must settle on one of a finite set of candidates x – readings of a sentence, participants for a role, antecedents for a pronoun – and each candidate carries a score $s(x)$. The Gibbs distribution at temperature T assigns the candidates probabilities

$$P_T(x) = e^{s(x)/T} / Z_T, \quad (2)$$

with Z_T the normalization over the candidate set. As $T \rightarrow 0$ the distribution piles all its mass on the best candidate: zero temperature is argmax, deterministic choice, and the odds against every non-optimal candidate *diverge* – which is, we contend, exactly the behavior linguists record with the ungrammaticality star. Acceptability is pervasively gradient and predictable from suitably normalized model probabilities (Lau, Clark, and Lappin, 2017), once probability is no longer conflated with frequency of attestation (Pereira, 2000); the categorical judgments of the tradition are recovered as the cold limit, the star marking a singularity of the odds rather than a Boolean feature of the string. Appendix A develops (2) from scratch and traces its career through simulated annealing, Hopfield networks, Boltzmann machines, Harmonic Grammar, and the transformer.

The cold limit is not merely an idealization of judgment data; it also describes familiar grammatical selection mechanisms. Pāṇini defines the instrument case by a superlative: the *karāṇa* is *sādhakatamam*, ‘the *most* effective means’ (Aṣṭādhyāyī 1.4.42), just as the *karman* is *īpsitatamam*, ‘the *most* desired’ (1.4.49). A superlative over candidate participants is an argmax; an argmax is the zero-temperature limit of a softmax; and a softmax over scored candidates is exactly what the attention mechanism computes (Appendix A.5). Kāraka assignment is a competition run cold: the candidates in (2) range over the clause’s participants, and the score is supplied by the proto-role entailments of

Dowty (1991), with the subject-selection hierarchy $A > I > O$ of Fillmore (1968) as the form book of expected winners. *The key opened the door* is the instrument winning in the absence of an agent; the classic star **John and the key opened the door* is an address clash – coordination distributes a single linking instruction over conjuncts that bind different addresses (CAUSE vs. INS) in the verb’s decomposition – zeugma, run cold.

Two points deserve emphasis before we turn the dial up. First, assignment runs cold *everywhere*: the competition must end in an argmax, which is why grammar is fast, automatic, and decisive in every language. Warmth enters not in the assignment of structure but in the evaluation of what has been assembled. Second, the claim of literalness. When *kāraka* assignment is construed as a softmax competition, no analogy is being drawn: attention *is* equation (2), the $1/\sqrt{d}$ divisor in its defining formula is a fixed temperature (Appendix A.5), and Pāṇini’s superlative picks out its zero-temperature limit. The linguist who has fitted a maxent grammar has already estimated the parameter; the physicist, the phonologist, and the transformer engineer are using one equation, and this paper merely declines to pretend otherwise.

3 Meat is murder

The copula contributes subsumption, and the parser’s work on (1a) and (1b) is identical: one routine, two assembled representations. Writing the readouts as contractions against the IS_A matrix E^0 of Kornai (2023),

$$\begin{aligned} (3) \quad & \text{a. } d_a = E_{ij}^0 \text{ snow}_i \text{ white}_j \\ & \text{b. } d_b = E_{ij}^0 \text{ meat}_i \text{ murder}_j \end{aligned}$$

everything depends on the temperature at which the readout is run. Retrieval against learned embeddings comes with a temperature parameter (Domingos, 2025): at $T = 0$ the softmax over similarities degenerates to exact match and reasoning is purely deductive – $d \approx 1$ for snow under white, and $d \approx 0$ for meat under murder in the taxonomy of any socialized adult, since that taxonomy does not place it there. We will argue that the second value is a learned artifact, and that the sentence is generically true in the lexicon the socialization overwrote.

Direct subsumption does not exhaust what runs at zero. The statics of *meat* contain, as preconditions, the production of the thing itself: meat is flesh taken for the table, and – humans not being carrion-eaters – taken means killed. *Meat requires killing* is therefore available deductively, with no borrowing at all, and *murder* decomposes, equally cold, as *killing* plus WRONG. Now run the chain: meat requires killing; were that killing murder, meat would require a wrong; and what can only be had through a wrong is itself wrong – the transmission principle is as cold as the rest. The slogan is thus deduction at every step but one, the step across the person/animal boundary in the victim slot of *murder*.

The restriction of victims to persons is not part of the naive entry: the child who learns where meat comes from reacts with distress before any rhetoric has reached them (Bray et al., 2016); some hold to that reaction against the practice of their own family, and on explicitly moral grounds (Hussar and Harris, 2010). The child is then *taught* the

restriction, the way one is taught that whales are not fishes. Two things are taught, in fact, and the slogans of Section 4 will need both: that the victim slot of *murder* admits persons only, and that killing for food falls on the legitimate side of a line that killing for other ends does not. (The features a dictionary lists for *murder* – agency, intent, unlawfulness, a victim who is a person – are these two lessons spelled out; unlawfulness is the legitimacy line by another name.) Before either lesson, *meat is murder* is true in the naive lexicon exactly as generics are true. The lexicon is conservative: an entry records the typical – *every donkey has four legs* is true by definition, unfalsifiable by the odd lame donkey – and entries are revised not by counterfactual pressure but by prevalence, the way Hungarian *kocsi* moved from ‘horse-drawn coach’ to ‘car’ when the motor variety became the prevalent one. Cold evaluation over the taught lexicon therefore fails the sentence at the two taught steps and at no other, and what the slogan requests is accordingly not a new crossing but the *suspension of a taught one*: as T rises, *slaughter* – adjacent to *murder* in every feature, since the naive entry never separated them – resumes the connection that socialization severed.

Why this step and no other, if the thermostat is one? Because the barriers differ. A single global T weakens every score gap at once, but it weakens them in proportion, and the gap at the taught restriction is the smallest in the chain: the naive entry never separated *slaughter* from *murder*, and the restriction is a penalty attached to one candidate after the fact, not a distance built into the statics. The deductive steps – meat requires killing, murder requires a wrong – are gaps of a different order, and stay effectively cold at any T a hearer will tolerate. As T rises the lowest barrier is crossed first. This is not a special mechanism but the reason annealing works at all (Appendix A.2), and it yields a prediction: the first thing to melt is the most recently taught exception. Herein lies the craft, and the economy: rhetoric is cheap exactly where convention had to work hard, for the suppressed reading is still resident in the statics, and a three-word slogan suffices to reactivate it. Nor is the boundary in question an arbitrary one to lean on. The line around the person – who counts as one for the purposes of *murder*, of *rights*, of *we* – is the most frequently redrawn line in the history of the lexicon, and every redrawing was, in its day, a prevalence-driven revision of a typicality default of the kind just described. A slogan that pushes on it is pushing where the lexicon has given way before.

Persuasion, on this account, is annealing in the metallurgist’s strict sense (Appendix A.2) – not the deposition of new material but the re-heating of a quenched arrangement so that it may resettle, the melt retained, or not, on cooling. Such revisions run in both directions and do cool into convention: the reconstrual that admits stone lions – clear negative cases of *lion* in any normal context – into the noun’s positive extension, forced by the non-vacuity of the combination, now runs as effortless routine for any material name paired with any concrete sortal (Kamp and Partee, 1995); there the crossing was taught rather than untaught, and the mechanism is one, prevalence-driven revision of typicality defaults, with the direction set case by case.

The rhetorical force of *meat is murder* is the engineered gap between the temperature its form promises and the temperature its content requires. The form is maximally cold: copula, generic subject, the very shape of taxonomy – the shape of *snow is white*, the tradition’s paradigm of analytic unexcitability. It is worth noticing that the paradigm

arrives pre-cooled. *Snow is white* entered the tradition as the stock atomic sentence of Tarski (1944), and the formalization that came with it types *snow* as a name and the sentence as an atomic predication, so that the genericity the pair turns on – *snow* and *meat* are mass nouns alike, and both sentences are bare copular generics in the sense of Carlson (1977) – is hidden by fiat before evaluation begins. A compositional derivation that inherits the paradigm will accordingly separate the pair on the subject side, for a reason that has nothing to do with meat or murder. The content verifies only warm. Neither side of this gap is a matter of felt coercion, on pain of circularity. The cold side has ordinary morphosyntax: bare or generic subject, individual-level present copula, gnomic rather than episodic anchoring. The warm side has geometry: the E^0 retrieval value of (3b) at $T = 0$, taxonomic distance, or the overlap of the two regions’ supports in a sparse-feature basis (Bricken et al., 2023). The hearer who runs the sentence cold gets falsity; the hearer who lets the neighborhoods touch gets the point; and the push one feels is the sentence daring its reader to warm up. Truth value, entailment, similarity, and uptake are not four analyses here but one vector read at four temperatures, the division of Section 1.

One thermostat, however, does not mean one setting. The situation is that of terrestrial climate: a single sun and one heat equation, and nonetheless tundra and tropics. The mechanism (2) is universal; its operating point is set locally, in layers. Assignment runs cold everywhere, as noted in Section 2. Evaluation has a conventional base climate: each language’s morphosyntax fixes which forms promise cold readings – genericity marking, gnomic aspect, obligatory mood are so many published thermostat settings – and register moves the setting wholesale, the mathematical register pinned near zero, legal drafting deliberately cold, poetry warm by convention. What a single utterance negotiates is the *weather*: a local departure from the climate its form announces. The engineered gap of (1b) is in this sense doubly determined – the copular generic is cold in the climate of English, and the slogan asks for a heat wave.

The borrowings, finally, are auditable. In a sparse-feature basis (Bricken et al., 2023; Templeton et al., 2024) the dimensions along which *meat* and *murder* overlap can be named and, if the hearer resists the rhetoric, disputed one by one; and the evaluation, unlike the parse, is deliberate multi-step inference, so the theory expects the parse of (1b) to survive ablation of the reportable workspace of Gurnee et al. (2026) while its rhetorical uptake does not. The prediction is stated here because it marks the difference between an analysis and an allegory: the temperature story is testable in an architecture whose internals can be recorded, decomposed, and perturbed.

4 Property is theft, taxation is theft

If the analysis of Section 3 is right, the three-word copular slogan should be a productive genre with a fixed anatomy: a form that promises cold, a content whose chain is deduction at every step but one, and a single taught boundary whose suspension the slogan requests. The genre obliges, and has the good grace to supply minimal pairs of its own:

- (4) a. Property is theft. (Proudhon, 1840)

b. Taxation is theft.

Theft decomposes as *taking* plus WRONG, exactly as *murder* decomposed as *killing* plus WRONG; and in both slogans the chain runs cold up to a single boundary, which in both cases is the taught line between legitimate and illegitimate taking. The child’s naive entry for taking does not distinguish the landlord’s rent from the pickpocket’s haul, or the tax collector’s demand from either; the distinctions are taught, entrenched, and – the slogans bet – suspendable. (4a) asks the hearer to warm away the taught legitimacy of accumulated holdings; (4b), of public levies. The forms are identical and maximally cold; the requested heat waves blow from opposite political quarters.

The pair is instructive on precisely this point. A theory that explained the force of (1b) by covert appeal to the analyst’s sympathy for its conclusion would face an awkward choice between (4a) and (4b); the temperature account faces none, because it measures the craft, not the merits. In each case the machinery reports the same facts: the gap between promised and required temperature, the location of the one warm step, and the residue of the naive entry that makes three words sufficient. Whether a given hearer’s taxonomy *should* resettle when annealed – whether the crossing ought to be retained on cooling – is a question the hearer’s other commitments decide, and the analysis has no vote. What the analysis does predict is where resistance will be voiced: at the taught boundary and nowhere else. Nobody disputes the slogans by denying that taxation involves taking, or that meat involves killing; the dispute goes straight to the victim slot and the legitimacy line, which is where the theory says the only warm step is.

5 Anaphora resolution run over discourse

We now run the whole machine on a case that the tradition would have to parcel out among syntax, semantics, and pragmatics (Kornai, 2012):

- (5) Someone disrespected his brother, so he beat him to pulp, because blood is thicker than water.

The grammar’s contribution is limited but precise: it supplies the index structure. In the first clause *someone* is linked as subject of *disrespect* and *his brother* as object, with *his* an addressing instruction that already admits two readings – the disrespector’s own brother, or a second party’s; write *p* for the disrespector, *q* for the brother-haver, and *b* for the brother. In the second clause *he* and *him* are addressing instructions pure and simple, *so* and *because* are relationals whose first slots want clause-sized matters, and the proverb is a bare copular generic. The grammar’s entire deliverable is thus *r* beat *s*, with *r* and *s* unresolved addresses – an addressing indeterminacy of exactly the kind familiar from PP-attachment – and syntax has nothing further to say. The candidates are not three individuals but the joint assignments to the triple (*his*, *he*, *him*). With *his* read as *p*’s own, the players are *p* and *b* and (*r*, *s*) ranges over the two orderings of {*p*, *b*}; with *his* read as a second party’s, *q* joins, and (*r*, *s*) ranges over the six ordered pairs of distinct members of {*p*, *q*, *b*}: eight assignments in all, and *q* is in the competition only under

the second reading. Yet every speaker of English resolves the addresses the same way: it is q, not b and not p, who did the beating, and p who was beaten.

The resolution runs entirely in the evaluation. The statics of *disrespect* contain ATTACK; the statics of *brother* contain FAMILY. The proverb is the interesting piece: run cold and literally it is true – the viscosity of blood is roughly four times that of water – and perfectly useless, since the BECAUSE-delta demands a matter that explains a beating, and hemodynamics explains no beatings. What the proverb actually contributes is its *annealed* reading, in the sense of Section 3: a boundary crossing that was warm for its first hearers has cooled into convention, and the phrase is now a lexicalized unit whose content is a bare comparative over the statics, *family er_ gen* – family outranks non-family, in the primitives of Kornai (2023). From there everything runs at zero temperature, and the chaining is E^0 contraction: disrespecting a brother is an attack on family; family outranks non-family, so an attack on family counts as an attack on self; and the sociobiological law ‘attack *should be* followed by counterattack’ – a two-state before/after schema with modal force, no different in kind from the lexical entry of *rest*, which also holds a *before(tired)* and an *after(has energy)* (Kornai, 2019) – delivers counterattack by the offended party. The eight assignments now face the competition (2) one more time: only $r = q$, $s = p$ lets the BECAUSE-delta discharge, so the argmax is unanimous, and it settles *his* in the same stroke – the brother must be q’s, or there is no offended party distinct from the attacker. Anaphora resolution is *kāraka* assignment run over the discourse: the same candidates, the same scores, the same cold limit.

Note what the example does *not* require. No rule ‘if someone attacks your brother you should counterattack’ is stored anywhere. The knowledge deployed is the deductive closure of a much smaller generating set: lexical statics and a few laws. No other arrangement could hold an open-ended repertoire of such cases. It also keeps the analysis explicit. The relevant inferences are attributed to lexical knowledge and a small number of general principles, not left as unexplained pragmatic residue.

The genre that industrializes this observation already exists. The original Winograd sentence (Winograd, 1972)

- (6) The city councilmen refused the demonstrators a permit because they {feared / advocated} violence.

delivers, in either variant, the same grammatical output: *they* is an unresolved address over {councilmen, demonstrators}. A single lexical swap flips the argmax, because the BECAUSE-delta demands a matter that explains a refusal, and the statics do the scoring – councils issue permits and fear disorder, demonstrators march and advocate. The Winograd schema (Levesque, Davis, and Morgenstein, 2012) is precisely the minimal-pair genre the present theory predicts should exist: delta structure identical across the pair, all the work in the evaluation, and no amount of syntactic sophistication sufficient to resolve it. The subsequent career of the schema challenge is equally predicted: constructed as a wall against statistical pattern-matching, it fell to large language models (Kocijan et al., 2023) – systems whose lexical statics are learned geometry and whose binding machinery is attention, that is, systems that resolve addresses by running exactly the competition (2) over exactly the resources the analysis above deploys. On the present account their

success on the schemas is not a confounding of the test but a confirmation of the theory of the test.

6 One thermostat

Four classical mysteries have now been put in the form of equation (2), and it is worth collecting them. In *demarcation* the candidates are competing analyses of a string, and the ungrammaticality star marks odds diverging in the cold limit. In *analyticity* the candidates are the neighboring sense-continuations of the subject, and the analytic truth is the retrieval that survives as $T \rightarrow 0$. In *kāraka assignment* – and, run over discourse, in anaphora resolution – the candidates are participants. In *rhetoric* the candidates are evaluation policies, thermostat settings, whose relative accessibility the utterance’s form manipulates: what the slogan negotiates is which policy the hearer runs, not the physics of the running. One equation, one thermostat; and the discipline cuts both ways, since whatever cannot be put in the form of (2) does not get to share it.

What we do not claim is worth stating with equal care. We do not claim that the temperature parameter explains where the scores come from – the lexical statics carry that load, here taken from Kornai (2023) but replaceable by any sufficiently articulated lexicon, including the learned kind (Bricken et al., 2023). We do not claim that current language models validate the analysis merely by containing softmaxes; the substantive predictions – that the parse of (1b) survives workspace ablation while its uptake does not, that resistance to a slogan localizes at its one warm step – are empirical, and stated so as to be falsifiable. And we do not claim novelty for the machinery, which is exactly the point: the linguist who has read Hayes and Wilson (2008) has already used it, and the Appendix, to which we turn now, makes this identification inescapable.

What we do claim can be held to the standard any cross-domain identity must meet: variables defined independently in each domain, a derivation in each, a measurable parameter, and predictions under intervention. The variables are the candidates and their scores, supplied domain by domain; the derivation is the free-energy trade-off of Appendix A.1, the same in each; the parameter is fitted by the phonologist, fixed by the architect, and, in evaluation, set by the instruction; and the predictions under intervention are the two of Section 3, ablation of the workspace and the location of resistance. What remains owed is the derivation of the scores themselves, and that debt belongs to the lexical statics, not to the thermostat.

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A Temperature for the working linguist

This appendix is a self-contained tour of the one piece of machinery the paper leans on throughout: the temperature parameter of equation (2). Nothing here is original; everything here is load-bearing. We assume comfort with vectors and matrices and with one-variable calculus, and nothing more – in particular, no multivariable calculus. Readers who know the material under its machine-learning names can skim for the linguistic anchors; readers who know it as Harmonic Grammar will find their own field waiting for them midway through.

A.1 One equation

Suppose a system must settle on one of a finite set of candidates x – readings of a sentence, participants for a role, continuations of a text – and each candidate carries a cost $E(x)$, its *energy*. Think of this as a weighted count of violations or other sources of accumulated badness. Statistical physics assigns the candidates the probabilities

$$P_T(x) = \frac{e^{-E(x)/T}}{Z_T}, \quad Z_T = \sum_{x'} e^{-E(x')/T}, \quad (7)$$

the *Gibbs* (or Boltzmann) *distribution* at temperature $T > 0$; the divisor Z_T merely makes the numbers sum to one. This is equation (2) with score $s = -E$: high score is low energy. Everything the paper does with temperature is contained in the behavior of (7) at its two extremes, and both follow from first-semester properties of the exponential.

Cold. Let E^* be the lowest energy on offer and divide numerator and denominator by $e^{-E^*/T}$. Every candidate’s weight becomes $e^{-(E(x)-E^*)/T}$, which as $T \rightarrow 0$ tends to 1 if $E(x) = E^*$ and to 0 otherwise: the distribution piles all its mass on the best candidate(s). Zero temperature is *argmax* – deterministic choice, no hedging, and the odds against any non-optimal candidate diverge, which is the sense in which Section 2 called the grammaticality star a *singularity* of the cold limit.

Warm. As $T \rightarrow \infty$ every exponent tends to 0, every weight to 1: the distribution flattens toward indifference. In between, T sets the exchange rate between decisiveness and open-mindedness – literally so, as the next observation shows.

Two candidates suffice to see the whole mechanism. If the choice is between a and b , (7) gives

$$P_T(a) = \frac{1}{1 + e^{-(E(b)-E(a))/T}}, \quad (8)$$

the *logistic* (sigmoid) curve in the energy difference: the workhorse function of both neural networks and categorical data analysis is nothing but the two-candidate Gibbs distribution, with T controlling how steeply the S-curve rises. And a single differentiation shows what the whole distribution optimizes. Give candidate a probability p and b probability $1 - p$, and define the *free energy* of the assignment as expected energy minus T times entropy:

$$F(p) = \underbrace{pE(a) + (1-p)E(b)}_{\text{expected energy}} + T \underbrace{(p \ln p + (1-p) \ln(1-p))}_{-\text{entropy}}. \quad (9)$$

Setting $F'(p) = E(a) - E(b) + T \ln \frac{p}{1-p}$ to zero recovers exactly (8): *the Gibbs distribution is the unique compromise between minimizing cost and keeping one's options open*, with T the price of an option. (The same statement holds for any number of candidates; the two-candidate case is the one that is easy to verify with the tools at hand.) This is also the sense, and the only sense, in which the paper calls the parameter *one*: in every domain below, physical, phonological, or computational, T is the multiplier on the entropy term, the exchange rate at which commitment is traded for open options, whatever the scores are made of and whatever mechanism produces them. Cold systems buy commitment; warm systems buy entropy. Every subsequent appearance of temperature – in annealing, in learning, in sampling, in rhetoric – is this one trade-off wearing different clothes.

A.2 Annealing

A metallurgist who wants a strong blade does not simply cool the metal; she cools it *slowly*, holding it warm long enough for the atoms to find a low-energy crystal arrangement, because fast cooling – quenching – freezes in whatever defects the atoms happened to be sitting in. Kirkpatrick, Gelatt, and Vecchi (1983) turned this into a general optimization method, *simulated annealing*: to minimize a complicated cost function, wander among candidates, accepting improvements always and deteriorations of size ΔE with probability $e^{-\Delta E/T}$ – the Gibbs weight again – while slowly lowering T . Warmth is what lets the search climb out of local minima, configurations better than all their neighbors but worse than the optimum; the schedule of cooling decides whether the system ends in the crystal or in a defect. The paper's account of persuasion (Section 3) is this picture applied to the hearer: the rhetorical move requests a temporary warming, the borrowed inference is adopted, and if it is retained on cooling – annealed into the hearer's taxonomy – it thereafter verifies cold. The stone lion is a fully cooled such crossing; *blood is thicker than water* in (5) another, old enough that its literal reading now requires effort to even notice.

A.3 Energy landscapes in networks

Hopfield (1982) showed that a network of binary units $x_i \in \{0, 1\}$ with symmetric connection weights w_{ij} behaves exactly like a physical system with energy

$$E(x) = -\frac{1}{2} \sum_{i,j} w_{ij} x_i x_j, \quad (10)$$

a quadratic form in the state vector. Let each unit repeatedly ask whether flipping itself would lower E , and the network settles into a minimum of the energy landscape; store patterns by setting the weights so that each pattern is such a minimum, and settling *is* content-addressable retrieval – present a corrupted fragment, and the dynamics complete it. But deterministic descent inherits the metallurgist's problem: it quenches, halting in whatever local minimum lies downhill. The *Boltzmann machine* of Ackley, Hinton, and Sejnowski (1985) is the annealed repair: let each unit turn on stochastically, with the logistic probability (8) applied, unit by unit, to the energy difference its own flip would make. By (10), turning unit i on lowers the energy by $\sum_j w_{ij} x_j$, its net input, so the

two-candidate rule becomes

$$P_T(x_i = 1) = \frac{1}{1 + e^{-\sum_j w_{ij}x_j/T}} \quad (8')$$

or, collecting the n rules into one vector equation,

$$p = \sigma\left(\frac{Wx}{T}\right), \quad p_i = P_T(x_i = 1 \mid x_{-i}), \quad (8'')$$

with σ the logistic function of (8) applied coordinatewise, W the weight matrix, and p the vector of each unit's conditional probability of turning on given the current state x_{-i} of the others: one temperature, n coupled S-curves. (A bias term, if wanted, is a unit clamped at 1.) A network of such units, run at a fixed T , samples the Gibbs distribution (7) over its global states; lowered on a schedule, T anneals it exactly as in Appendix A.2, the warmth letting the network climb out of the local minima of (10) in which deterministic descent would have halted. Its learning rule adjusts the weights so that the statistics the network exhibits when running free match the statistics of the data it is shown – the first working demonstration that warm sampling is not just a search trick but a medium for *learning*.

A.4 The linguist has seen this before

Rebrand energy as its negative and it has been in linguistics for decades: Smolensky (1986) called $-E$ *harmony*, and Harmonic Grammar scores a candidate analysis by the weighted sum of its constraint violations – precisely an energy function, linear in the violation counts. Optimality Theory (Prince and Smolensky, 1993) is its cold, strictly-ranked limit: let each constraint's weight grow so large that no amount of success on lower constraints can compensate a single violation of a higher one, and weighted trade-offs collapse into lexicographic domination – the tableau is a frozen Gibbs table, showing only the argmax. And maximum-entropy phonotactics (Hayes and Wilson, 2008) *is* the Gibbs distribution (7), by that name: a probability over candidates proportional to the exponential of their weighted violation score, fitted to corpus data, gradient acceptability and all. (The fitted scale is a convention: doubling every weight and T together changes nothing, so a maxent grammar fixes T only relative to its weights, and it is that ratio the data determine. The limit is exact for any finite set of optima, which some weighting always reproduces (Prince and Smolensky, 1993, p. 236), and fails only where one constraint's satisfaction can cost unboundedly many violations of another; Pater (2009) maps the boundary.) The path this appendix walks – from energy minimization through warm sampling to graded judgment – was thus walked by phonology a generation ago, in the opposite direction and under other names. What the paper adds is only the claim that the thermostat is *one*: the parameter these models fit for phonotactic gradience is the same dial that Sections 2–5 turn for *kāraka* assignment, analyticity, anaphora, and rhetoric.

A.5 From energies to scores: softmax and attention

Machine learning writes $s = -E$ and calls (7) the *softmax* of the score vector at temperature T . Two places the reader has already met it. First, the sampling knob: every deployed language model turns its next-token scores into probabilities by softmax, and the user-facing “temperature” setting is T itself – $T < 1$ sharpens the distribution toward the argmax (deterministic, repetitive, safe), $T > 1$ flattens it (creative, risky), $T = 0$ is greedy decoding. This knob sits on the output softmax, downstream of everything the network has computed: what it rescales is the choice of wording for a reading already settled inside. Second, and closer to this paper’s machinery: an attention head compares a query vector q against key vectors $k_1, \dots, k_n$ by dot products and turns the resulting scores into weights by softmax (Vaswani et al., 2017) – a Gibbs competition over which positions to read. The celebrated $1/\sqrt{d}$ divisor in the attention formula is a *fixed temperature*: the dot product of two d -dimensional vectors with independent, unit-variance coordinates has variance d , so dividing by $\sqrt{d}$ restores unit scale – the temperature is set once, by the architecture, so that the competition starts neither frozen nor boiling. When Section 2 construes *kāraka* assignment as a competition run cold, the claim is thus not a metaphor imported from physics but a description of the literal mechanism: attention *is* (7), and Pāṇini’s superlative picks out its zero-temperature limit.

A.6 Temperature in learning: regularization

The free-energy trade-off (9) reappears wherever a system must *learn* rather than merely choose, because a learner faces the metallurgist’s dilemma in time: commit too hard, too early, to the training data and it freezes in the data’s accidents – the failure called *overfitting*, which is quenching by another name. *Regularization* is the collective name for the devices that keep a learner warm, and they come in two grades. Three are temperatures in the literal sense of (7). Label smoothing replaces the training targets 0 and 1 by softened values, warming the target distribution. Knowledge distillation (Hinton, Vinyals, and Dean, 2015) runs a trained teacher’s softmax *warm* (T of 2 to 5) so that the near-miss structure of its judgments – which wrong answers are almost right – becomes visible in the targets a smaller student learns from; the information carried in the warm distribution and destroyed by the argmax is precisely what the tutorial’s cold limit discards. And after training, *temperature scaling* fits a single post-hoc T so that the model’s confidence matches its accuracy: modern practice literally installs a thermostat between the model and its users. The rest are warmth by analogy, and the appendix claims no more for them. Weight decay adds $\lambda \sum_i w_i^2$ to the cost, a standing refusal to let any weight commit to infinity on finite evidence. Dropout injects noise by silencing random units during training, so that no analysis can depend on one fragile configuration – perturbation in the role that annealing gives to uphill moves. Early stopping halts the cooling schedule before the quench. The slogan form of all this: *learn warm, serve cold* – and the reader who has followed the paper will recognize the same layering in Section 3’s climate and weather, conventionally fixed operating points with locally negotiated departures.

A.7 Back to the main text

Here we provide a reverse index of the paper’s uses of temperature, mapped to their tutorial homes. The ungrammaticality *star* is the divergence of odds in the cold limit of (7) (Section 2); *kāraka assignment*. Anaphora resolution is the attention competition of Section A.5 run at $T \rightarrow 0$ (Sections 2 and 5); the *analytic truth* is the retrieval that survives cooling; *rhetoric* is temperature negotiation, and *persuasion* is annealing (Section A.2), with the stone lion and the blood proverb as cooled crossings; *climate* is the conventionalized schedule – the per-language, per-register operating point – and *weather* its utterance-level negotiation, the same division that machine learning draws between the trained thermostat and the sampling knob. Finally, the *learner’s* need to stay warm (Section A.6) is why none of the cold behavior is innate rigidity: the grammar is cold because it has finished cooling, not because it never had a temperature.

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